

Samik Banerjee, Ph.D.

AI Scientist | Computer Vision & Deep Learning Expert

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Cold Spring Harbor Laboratory, New York, USA

Professional Summary

Computer Scientist (Ph.D.) specializing in **Computer Vision** and **Generative AI**. Expert in architecting **scalable designs** and novel **segmentation** and **registration** algorithms for **petabyte-scale** datasets and deploying production-grade pipelines for medical imaging and computational neuroscience. Proven track record of delivering end-to-end software tools for global research collaborations through **Cross-Functional Leadership**. Proficient in **PyTorch** and **TensorFlow**, with a strong track record of publishing in **top-tier journals and conferences** (Nature, Nature Machine Intelligence, Nature Communications, ECCVW, CVPRW, etc.), while serving as a Reviewer for multiple top-tier journals.

Technical Skills

- **Generative AI & Deep Learning:** PyTorch, TensorFlow, Siamese Networks, CNNs, GANs, Scikit-learn, Model Optimization, Inference Acceleration.
- **Computer Vision:** Segmentation, Object Detection, Skeletonization, Volumetric Registration, 3D Reconstruction.
- **Languages:** Python, C++, SQL, MATLAB, JavaScript (Node.js), TypeScript (Angular).
- **Cloud & HPC:** AWS (EC2, Batch), Docker, Distributed Computing, HPC.
- **Tools & Libraries:** OpenCV, ImageJ, Neuroglancer, Pandas, NumPy, Git.

Professional Experience

Cold Spring Harbor Laboratory (CSHL) | New York, USA | Computational Science Analyst | Sep 2018 – Apr 2026 | Individual Research Consultant (Remote) | Apr 2026 – Present

- **Scalable AI Pipelines:** Engineered automated pipelines to detect mammalian neuronal connectivity, processing **PB-scale** whole-brain imaging data. Reduced analysis time from months to days.
- **3D Atlas Generation:** Led the computational development of the **first cell census atlas** of the mammalian primary motor cortex (*Nature*, 2021), utilizing deep learning for precise cell counting and registration.
- **Algorithm Development:** Formulated “SemSeg,” a semantic segmentation framework achieving **state-of-the-art accuracy** in identifying neuroanatomical structures in microscopic data (*Nature Machine Intelligence*, 2020).
- **Full-Stack Tooling:** Developed an interactive web-based visualization platform (Angular/Node.js) used by international collaborators to visualize **mammalian data (mouse, marmoset, human)** at **sub-micron resolution**, and incorporated a platform to annotate and view 3D single-cell data on AWS.
- **Multi-omics Analysis:** Spearheaded integration of transcriptomic (RNA-seq) and genetic barcode data to map cell-type specific connections, integrating molecular data with anatomical imaging.
- **Deployment:** Deployed analysis workflows on **AWS EC2** clusters, ensuring high availability and scalability for multi-institutional projects (BICCN, BICAN, BRAIN CONNECTS).

Indian Institute of Technology (IIT) Madras | Chennai, India | Doctoral Researcher | Jul 2012 – Sep 2019

- **Generative AI for Restoration:** Designed **LR-GAN** to hallucinate high-resolution faces and **SD-GAN** to recover facial parts from **occluded images**, utilizing structural constraint losses on surveillance-grade inputs.
- **Pose-Invariant Generation:** Developed **PosIX-GAN** to generate multi-view face images from single monocular inputs, handling extreme pose variations.
- **Cross-Domain Adaptation:** Pioneered **Transfer-CNN**, an unsupervised Domain Adaptation technique using **Mutual Variation of Information** and Eigen-domain transformation to align surveillance face data with gallery images, boosting recognition rates by **15%**.
- **Cosmetic Invariance:** Engineered **MakeUpMirror**, a Siamese CNN architecture to verify faces across heavy cosmetic changes, learning makeup-invariant features.
- **Surveillance Systems:** Deployed robust object detection and tracking algorithms for identifying targets in satellite imagery (DRDO project) and real-time video feeds.

Heritage Institute of Technology | Kolkata, India | Assistant Professor (CSE) | Sep 2010 – Jul 2012

- Directed research for 50+ students and taught advanced courses in Algorithm Design and Computer Graphics.
- **Early Research:** Orchestrated M.Tech research projects on **Quantum Cryptography**, **Reversible Logic**, and Error-Detection circuits.

Selected Publications & Manuscripts

1. “Semantic segmentation of microscopic neuroanatomical data by combining topological priors with encoder-decoder deep networks.” *Nature Machine Intelligence*, 2020.
2. “A guide to the BRAIN Initiative Cell Census Network data ecosystem.” *PLoS Biology*, 2023.

3. “A multimodal cell census and atlas of the mammalian primary motor cortex.” *Nature*, 2021.
4. “Cellular Anatomy of the Mouse Primary Motor Cortex.” *Nature*, 2021.
5. “Solving the where problem and quantifying geometric variation in neuroanatomy using generative diffeomorphic mapping.” *Nature Communications*, 2025.
6. “PosIX-GAN: Generating multiple poses using GAN for Pose-Invariant Face Recognition.” *ECCV Workshop*, 2018.
7. “LR-GAN for Degraded Face Recognition.” *Pattern Recognition Letters*, 2018.
8. “Mutual variation of Information on Transfer-CNN for Face Recognition with degraded probe samples.” *Neurocomputing*, 2018.
9. “MakeUpMirror: Mirroring make-ups and verifying faces post make-up.” *IET Biometrics*, 2018.
10. “Soft-Margin Learning for Multiple Feature-Kernel Combinations With Domain Adaptation.” *CVPR Workshop on Biometrics*, 2016.
11. **(Under Review)** “Skeletonization of neuronal processes using Discrete Morse techniques from computational topology.” *arXiv preprint arXiv:2505.07754*. May, 2025. (Under Review in *Nature Communications*)
12. **(Under Review)** “3D multimodal histological atlas and coordinate framework for the mouse brain and head.” *bioRxiv 650246*. May, 2025. (Under Review in *Nature Neuroscience*)
13. **(Under Review)** “A three-dimensional histological cell atlas of the developing human brain.” *bioRxiv 628811*. December, 2024. (Under Review in *Nature*)
14. **(Under Review)** “Survey of loss functions for training deep-learning models in computer vision.” *SSRN*, 2025. (Under Review in *CVIU, Elsevier*)
15. **(Preprint)** “SD-GAN: Structural and Denoising GAN reveals facial parts under occlusion.” *arXiv preprint arXiv:2002.08448*, 2020.

Education

- **Ph.D. in Computer Science & Engineering** | IIT Madras, India | 2019
 - *Thesis*: “Face Recognition from Degraded Images using Statistical and Deep Learning Techniques” (Computer Vision & Deep Learning).
- **M.E. in Computer Science & Technology** | BESU Shibpur, India | 2010
 - *Thesis*: “Secure Scan Design” (Network Security & Cryptography).
- **B.Tech. in Computer Science & Engineering** | St. Thomas’ College of Engineering & Technology, India | 2008
 - *Thesis*: “A Robust Group Authentication System with Multi-level Security” (Cryptography & Network Security).

Research Sponsors

- **National Institutes of Health (NIH), USA**: BRAIN Initiative, NINDS, NIMH.
- **Government of India**: DRDO (Defense Research), MIMP, Office of Principal Scientific Adviser.

Selected Conferences, Meetings & Workshops Attended

- **Neuroscience & Computational Brain Research**: Society for Neuroscience (2018, ’22, ’23); BRAIN Initiative Meetings (2019, ’23); WWCBR (2017, ’18, ’20 - *Invited Speaker*).
- **Computer Vision & Machine Intelligence**: ICVGIP (2014, ’16); PReMI (2017); PerMIn (2015); NCVPRIPG (2013, ’23).
- **Industry & Technical Workshops**: NVIDIA Developer Connect (2017); GPU Technology Conference (2016); Workshop on Digital Video Analytics (2012).

Awards & Achievements

- **FlyTxt Fellowship Award**, PReMI Conference (2017).
- **Pan-India Winner**, Samsung Challenge on SMS Classification (2017): Developed an NLP/SVM-based classifier.
- **Regional Winner**, Microsoft Code.Fun.Do Hackathon (2014, 2016):
 - 2016: “Cookey” - Windows App for inventory management.
 - 2014: “EYE-YAAR” - Computer Vision app for eye check-up using **Kinect Depth Sensors**.
- **Star-TA Award**, IIT Madras (2017).
- **Top 10% Oral Presentation**, ICVGIP (2014).